

Cover Letter — Nature Machine Intelligence (Analysis)

Manuscript: Methodological choices can outweigh algorithmic choices in transcriptomic immunotherapy prediction **Author:** Zhixian Huang (sole author), Putian University — huangzhixian324@gmail.com **Format requested:** Analysis (~2,760 words of main text excluding Methods; 3 figures + 3 tables)

Dear Editors,

I am pleased to submit the above manuscript for consideration in your Analysis format.

Machine learning models that predict treatment response from transcriptomics are proliferating in oncology, and immunotherapy is the fastest-growing application. The evaluation literature, however, keeps producing contradictory rankings: recent cross-cohort evaluations of nine predictors and twenty signatures, a 1,133-patient cross-cancer meta-benchmark, and a 35-task machine-learning benchmark all hold the benchmarking methodology fixed and compare algorithms. None asks the question on which the credibility of this model class depends: how much of a benchmark's outcome is determined by the benchmarking design itself?

The manuscript answers that question with a controlled design that holds six established prediction methods, the patients, and the expression data fixed, and varies only the benchmarking design. Three findings:

1. Response definition outweighs algorithm choice. Switching a molecular surrogate endpoint to a clinical endpoint (RECIST v1.1) in the same patients collapses every method whose features overlap the endpoint-defining genes, with a point-estimate effect $3.5\times$ the largest between-method gap. The collapse replicates in executed external cohorts spanning five cancer types and is specific to gene-defined surrogates, not to clinical-definition disagreement.
2. The circular construction is instantiated practice, not a hypothetical. A structured field audit identifies seven published studies that train or select immunotherapy-response predictors against gene-expression-derived labels in cohorts without ICI outcomes. The design rules proposed here therefore bind this literature segment directly.
3. Statistical floors are quantifiable. A complete 36-cell permutation-test matrix with family-sensitivity analysis shows that no trainable method is distinguishable from label shuffling on any clinical endpoint at $n \leq 43$, with the first detectable signal at $n = 73$, matching an independently anchored power analysis.

The contribution to machine-learning practice is the benchmarking methodology itself: a design-choice decomposition that transfers to any omics prediction benchmark; a reporting convention whose binding rule is the declaration of endpoint-defining genes, with the Circularity Detection Score shipped as an installable package (v2.0.0, validated across eight real endpoints); and a

minimum-standards checklist derived from power analysis. All code, data, and per-cell outputs are open, Docker-reproducible, and publicly archived (Zenodo DOI: 10.5281/zenodo.22746254; GitHub, MIT). The Software Submission Checklist accompanies this submission because the central code is a core deliverable.

Why Nature Machine Intelligence. Your 2025 Analysis "Benchmarking AI-powered docking methods from the perspective of virtual screening" (Gu et al., Nat Mach Intell 7:509–520) established that systematic AI-method benchmarks with an evaluation-practice critique belong in this journal's Analysis format. The present manuscript applies the same template to transcriptomic immunotherapy prediction, where the benchmarking-design problem is documented rather than hypothetical. The design-choice decomposition, the circularity diagnostic, and the minimum-standards checklist transfer directly to any omics prediction benchmark and should interest your ML methodology readership beyond oncology.

Two candid notes. The clinical application is immunotherapy rather than basic biology; the methodological contribution, not the clinical setting, is what I believe fits your journal. And the study is single-authored; I know a benchmark asking for community trust has to earn it the hard way, which is why every analysis ran under archived, versioned protocols with a complete re-derivation audit trail.

This manuscript is original, is not under consideration elsewhere, and the sole author has approved the submission. I have no competing interests and no funding to declare. I have not discussed this work with any Nature Machine Intelligence editor prior to this submission.

Whether the work belongs here is your judgment alone; I will be grateful for it either way.

Sincerely, Zhixian Huang Putian University huangzhixian324@gmail.com